Polynomial System Solving via Factor Branching

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ABSTRACT

Polynomial systems are a fundamental language for describing algebraic and geometric problems in mathematics and its applications. Beyond finding individual solutions, one often seeks global properties of the solution space, such as its dimension and irreducible components, or structural properties and decompositions of the underlying polynomial ideal. For large polynomial systems with many variables and high degrees, determining these global properties can become intractable. Leading approaches include variants of Buchberger's algorithm and resultants. However, even the most modern approaches still often fail on practical problems or slight perturbations of standard benchmarks. Here we introduce a framework that divides a polynomial system into simpler branches by interreducing and factoring its equations. The framework only assumes that the solver used on each branch describes satisfiable (SAT) branches and certifies unsatisfiable (UNSAT) ones. Its variants return the minimal primes and a minimal primary decomposition of the ideal, and we prove every variant exact. In our experiments, we use Gröbner-basis solvers. On 20 generated and 31 size-graded benchmark systems, factor branching completes 19 and 27 within 60 seconds, against at most 16 and 20 for the baselines. On a system arising from research in quantum field theory, all nine baselines time out, whereas factor branching finishes in under a second and its primary variant computes a primary decomposition in under five minutes. Together, these ideas offer a path toward the exact analysis of previously inaccessible polynomial systems.

1. Introduction

Systems of multivariate polynomial equations appear whenever a symbolic model contains nonlinear algebraic constraints. Applications include robotics, coding theory, optimization, statistics, and physics (Sturmfels, 2002). In many of these problems, an isolated solution is not enough. One needs the dimension of the solution set, its irreducible components, or the exact relations that hold on every solution. These are questions about the ideal I generated by the equations and its variety $V(I)$, rather than about a single point (Cox et al., 2015; Kreuzer and Robbiano, 2000).

The outputs of algebraic algorithms vary along two axes (Figure 1):

- How far the output breaks down the ideal: it treats I as one piece, decomposes it into ideals that need not be prime or primary, or decomposes it into its prime or primary components.
- How much information the output retains: whether $V(I)$ is empty, properties of $V(I)$ such as its dimension, $\sqrt{I}$, or I itself.

Elimination by resultants provided an early systematic way to attack polynomial systems, but the introduction of Gröbner bases gave a common algorithmic framework for elimination, ideal membership, dimension, and related tasks (Cox et al., 2015; Kreuzer and Robbiano, 2000). Buchberger developed this framework in his 1965 thesis (Buchberger, 2006). Later algorithms changed how the expensive reductions are organized. F4 replaces individual polynomial reductions with sparse linear algebra, while F5 uses signatures to avoid many reductions to zero (Faugère, 1999, 2002). Primary-decomposition algorithms use Gröbner bases and factorization to separate components (Gianni et al., 1988). Factorizing Gröbner-basis methods split on individual polynomial factorizations and preserve the solution set; earlier methods also interreduce and prune redundant branches (Decker et al., 2025; Gräbe, 1995). Numerical algebraic geometry takes a different route: homotopy continuation computes isolated roots numerically, while witness sets represent positive-dimensional components (Sommese et al., 2001; Sommese and Wampler, 2005).

Recent work has also explored machine-learning methods for polynomial systems. Peifer et al. trained a reinforcement-learning policy for pair selection in Buchberger's algorithm (Peifer et al., 2020). Kera et al. trained

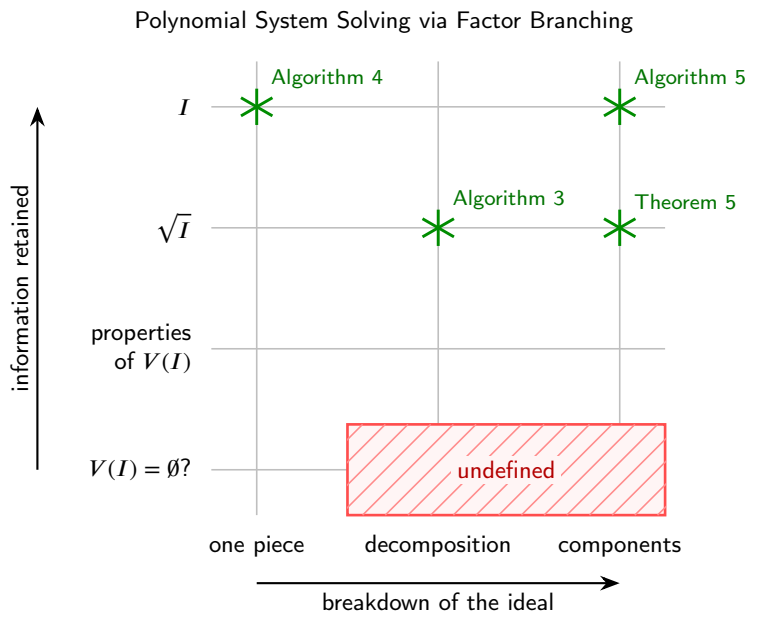


Figure 1: Outputs of polynomial-system algorithms on two axes. Columns order the outputs by how far they break down the ideal I, and rows by how much information they retain. Crosses mark the methods of this paper. The hatched cells are undefined, since whether $V(I)$ is empty is a single answer with nothing to break down.

transformers to compute Gröbner bases directly for a restricted class of zero-dimensional radical ideals (Kera et al., 2024). Yet these works are limited to small and specific polynomial ideals.

Despite this progress, large systems remain inaccessible, such as a quantum-gravity system from our research on which every standalone method in our comparison times out (Subsection 5.4).

Here, we introduce solver-agnostic methods for several of these outputs, all built from the same steps. Each method factors the equations and enumerates the minimal hitting sets of the resulting factor sets. A minimal hitting set contains at least one factor of every equation, so it defines a branch, a smaller system whose solutions are solutions of the original one, and $V(I)$ is the union of their solution sets. Interreducing and factoring a branch can split it further, and the methods repeat the split until it changes nothing. Each method then calls an exact solver on the branches and combines the results into its output. Unlike factorizing Gröbner-basis methods, which split on one factorization at a time inside a Gröbner-basis computation, the methods split on all equations at once before calling the solver, so any exact solver can serve. Since the worst-case complexity of computing a Gröbner basis is doubly exponential in the number of variables (Mayr and Meyer, 1982), breaking down a large system into smaller branches can be substantially faster than solving the original system directly.

Figure 1 places the methods of this paper on the two axes. The methods differ in the branch solver and in how they combine the branches. Factor branching asks the solver to describe each branch, including any positive-dimensional families, or to certify that it is UNSAT, and so returns a decomposition of $V(I)$ whose pieces may overlap (Section 3). Prime factor branching asks for the minimal primes of each branch and returns those of I, which give the irreducible components of $V(I)$ over K and their dimensions (Theorem 5). The remaining methods keep the exponents of the factors, and with them I itself. Ideal reduction uses the SAT branches to replace generators by smaller members of I (Subsection 4.1). Primary factor branching obtains the isolated primary components of I from its minimal primes and runs a general primary decomposition only when I has embedded components (Subsection 4.2).

We prove that every method is exact. Refinement terminates and preserves the solution set, ideal reduction preserves I, and prime and primary factor branching return the minimal primes and a minimal primary decomposition of I. Every numbered result is also formalized in Lean 4 with Mathlib, in the repository listed under Data availability. On generated systems with factor structure and on size-graded families from SymbolicData and from applications, factor branching with the best of its backends completes 19 of 20 and 27 of 31 systems within 60 seconds, against at most 16 and 20 for the direct Gröbner-basis and decomposition methods. On a system from quantum field theory, all nine standalone baselines time out, whereas factor branching with the `facstd` backend completes in under a second. Primary factor

branching computes the primary decomposition of this system in under five minutes, while Singular's `primdecSY` does not finish within 30 minutes on the original equations.

2. Preliminaries and notation

2.1. Polynomial systems, ideals, and varieties

Let K be a field, let $\overline{K}$ be its algebraic closure, and write $R = K[x_1, \dots, x_n]$. The input is a finite system $P = \{p_1, \dots, p_m\} \subseteq R$ of nonzero polynomials. Unless stated otherwise, a solution is a point in $\overline{K}^n$. For a finite set $Q \subseteq R$, its solution set is

$$V(Q) = \{a \in \overline{K}^n : q(a) = 0 \text{ for every } q \in Q\}.$$

We call Q satisfiable (SAT) if $V(Q) \neq \emptyset$ and unsatisfiable (UNSAT) if $V(Q) = \emptyset$.

We write $\langle Q \rangle$ for the ideal generated by Q.

For $X \subseteq \overline{K}^n$, its vanishing ideal $I(X)$ contains every polynomial in R that vanishes on all of X. The radical of an ideal J is $\sqrt{J} = \{g \in R : g^r \in J \text{ for some } r \geq 1\}$. Hilbert's Nullstellensatz gives

$$I(V(Q)) = \sqrt{\langle Q \rangle}.$$

Thus $V(Q) = \emptyset$ exactly when $\langle Q \rangle = \langle 1 \rangle$ (Cox et al., 2015; Kreuzer and Robbiano, 2000).

2.2. Saturation and decompositions

The saturation of an ideal J by $u \in R$ (Cox et al., 2015, Ch. 4, §4) is

$$J : u^\infty = \{g \in R : u^m g \in J \text{ for some } m \geq 0\}.$$

An ideal $\mathfrak{q}$ is primary if $gh \in \mathfrak{q}$ implies $g \in \mathfrak{q}$ or $h^m \in \mathfrak{q}$ for some $m \geq 1$, and then $\sqrt{\mathfrak{q}}$ is prime. Every ideal J of R has a minimal primary decomposition (Cox et al., 2015, Ch. 4, §8)

$$J = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_t$$

into primary ideals with distinct radicals $\mathfrak{p}_j = \sqrt{\mathfrak{q}_j}$, where no $\mathfrak{q}_j$ contains the intersection of the others. The radicals are the associated primes of J, and the minimal ones among them are the minimal primes of J. The minimal primes intersect to $\sqrt{J}$ (Cox et al., 2015, Ch. 4, §6). The components whose radicals are minimal primes, called isolated, are unique; the other components, called embedded, are not (Cox et al., 2015, Ch. 4, §8). Saturation keeps the components whose radicals do not contain u, $J : u^\infty = \bigcap_{u \notin \mathfrak{p}_j} \mathfrak{q}_j$ (Atiyah and Macdonald, 1969, Prop. 4.9). A prime that contains the intersection of finitely many ideals contains one of them (Atiyah and Macdonald, 1969, Prop. 1.11).

2.3. Monomial orders and Gröbner bases

Let $\mathbb{M}_n = \{x^\alpha : \alpha \in \mathbb{N}^n\}$ be the monomials of R. A monomial order $\prec$ is a total order on $\mathbb{M}_n$ such that every nonempty subset has a least element and

$$u \prec v \quad \Longrightarrow \quad uw \prec vw \qquad (u, v, w \in \mathbb{M}_n).$$

For a nonzero polynomial p, the leading monomial $\mathrm{LM}(p)$ is the greatest monomial of p under $\prec$, $\mathrm{LC}(p)$ is its coefficient, and $\mathrm{LT}(p)$ is its leading term.

For an ideal J, define its leading-monomial ideal by

$$\mathrm{LM}(J) = \langle \mathrm{LM}(f) : f \in J \setminus \{0\} \rangle.$$

A finite set $G \subseteq J$ is a Gröbner basis of J if

$$\mathrm{LM}(J) = \langle \mathrm{LM}(g) : g \in G \rangle.$$

A finite set Q is *interreduced* if every element is monic and no term of one element is divisible by the leading monomial of another. Algorithm 1 interreduces a finite set of nonzero polynomials.

Algorithm 1 Interreduction

Require: A finite set $Q \subseteq R \setminus \{0\}$ and a monomial order $\prec$
Ensure: An interreduced set that generates $\langle Q \rangle$
while there exist distinct $p, q \in Q$ and a term cu of p with $\mathrm{LM}(q) \mid u$ **do**
  $r \leftarrow p - \dfrac{cu}{\mathrm{LT}(q)} q$
  $Q \leftarrow Q \setminus \{p\}$
  if $r \neq 0$ **then**
    $Q \leftarrow Q \cup \{r\}$
  end if
end while
$Q \leftarrow \{p/\mathrm{LC}(p) : p \in Q\}$
return Q

The loop terminates. Let $\mu(Q)$ be the multiset of the monomials of all elements of Q. An iteration replaces the monomials of p by those of r, which lack u and are monomials of p or strictly smaller than u. If $r = 0$ or r already lies in Q, the iteration only removes the monomials of p. Either way, $\mu(Q)$ decreases in the multiset extension of $\prec$, the order generated by replacing an element of a multiset with finitely many smaller ones. It is well founded because $\prec$ is (Dershowitz and Manna, 1979). Interreduction preserves $\langle Q \rangle$ and $V(Q)$ (Cox et al., 2015, Ch. 2, §7), and it does not shrink the ideal generated by the leading monomials: an iteration with $u \neq \mathrm{LM}(p)$ keeps $\mathrm{LM}(p)$ as the leading monomial of r, and if $u = \mathrm{LM}(p)$, then $\mathrm{LM}(q)$ divides it and q stays in Q.

For a fixed monomial order, every ideal J has a unique interreduced Gröbner basis, its reduced Gröbner basis. The reduced basis is $\{1\}$ if and only if $J = \langle 1 \rangle$. Therefore, a Gröbner basis computation can describe a branch and detect when it is UNSAT (Cox et al., 2015; Kreuzer and Robbiano, 2000).

Reduction also tests containment. A polynomial g reduces to zero modulo a finite set G if repeatedly replacing g by $g - \frac{cu}{\mathrm{LT}(q)} q$, for a term cu of g and $q \in G$ with $\mathrm{LM}(q) \mid u$, reaches 0. If every element of a finite set G' reduces to zero modulo G, then $\langle G' \rangle \subseteq \langle G \rangle$, and hence $V(G) \subseteq V(G')$. If G is a Gröbner basis, the test is exact: $\langle G' \rangle \subseteq \langle G \rangle$ if and only if every element of G' reduces to zero modulo G (Cox et al., 2015, Ch. 2, §§3, 6).

2.4. Factor sets and hitting sets

Fix a factorization over K of each polynomial q of a finite system $Q \subseteq R \setminus \{0\}$:

$$q = c_q \prod_{f \in C_q} f^{e_{q,f}},$$

where $c_q \in K \setminus \{0\}$, the factor set C_q contains the distinct monic, nonconstant irreducible factors of q, and $e_{q,f} \geq 1$. We write $C(Q) = \bigcup_{q \in Q} C_q$. A point satisfies $q = 0$ exactly when at least one factor in C_q vanishes at that point.

A hitting set of Q is a set $H \subseteq C(Q)$ such that $H \cap C_q \neq \emptyset$ for every $q \in Q$. It is minimal if no proper subset is also a hitting set (Berge, 1989). We write $\mathcal{H}(Q)$ for the family of minimal hitting sets of Q. Each $H \in \mathcal{H}(Q)$ defines a branch in which all factors in H vanish.

3. Factor branching framework

This section we present a method that breaks a polynomial system P into simpler branches, from which a solver describes $V(P)$ and $\sqrt{\langle P \rangle}$ more easily than from P itself.

3.1. Framework description

Algorithm 2 breaks the system P into smaller branches by factoring each generator and enumerating the minimal hitting sets of the factor sets. Each hitting set generates a new branch, which we interreduce and factor again, until it reaches a fixed point. We call a branch *terminal* if one round leaves it unchanged.

Hitting sets are minimal only among the children of one branch, so the loop of Algorithm 2 can end with distinct branches H and H' such that $V(H) \subseteq V(H')$. Solving H then adds no solutions. PRUNE(W) returns the branches whose solution sets it cannot certify to be contained in each other.

Algorithm 2 Refinement

Require: A finite polynomial system P
Ensure: The terminal branches of P

$W \leftarrow \mathcal{H}(P)$
repeat
  $W_{\text{prev}} \leftarrow W, \quad W \leftarrow \emptyset$
  for all $Q \in W_{\text{prev}}$ **do**
    $Q \leftarrow \textsc{Interreduce}(Q)$
    $W \leftarrow W \cup \mathcal{H}(Q)$
  end for
until $W = W_{\text{prev}}$
$W \leftarrow \textsc{Prune}(W)$ ▷ Optional
return W

Interreducing P first can expose factors that its generators hide, but interreducing can also eliminate factorizations without exposing new ones. So whether to interreduce first depends on P.

Let $\mathcal{L}(P)$ denote the terminal branches returned by Algorithm 2 on P. For each $H \in \mathcal{L}(P)$, the branch solver must either describe $V(H)$ exactly or certify that $V(H) = \emptyset$. We assume that every solver in this paper terminates and returns a correct output.

Algorithm 3 Factor branching

Require: A set of polynomials $P = \{p_1, \dots, p_m\}$ and a branch solver $\mathcal{B}$
Ensure: A decomposition of $V(P)$

$S \leftarrow \emptyset$ ▷ Solution-set representations
for all $H \in \textsc{Refine}(P)$ **do**
  $S \leftarrow S \cup \mathcal{B}(H)$
end for
return S

A solved branch can prune the later branches whose solution sets it contains. Branches with fewer generators should then come first: they are likely to be solved faster and, being likely of higher dimension, to contain later branches.

3.2. Theoretical analysis

Theorem 1 (One-round branch decomposition). *For every finite $Q \subseteq R \setminus \{0\}$,*

$$V(Q) = \bigcup_{H \in \mathcal{H}(Q)} V(H).$$

Proof. Let $a \in V(Q)$. Every $q \in Q$ vanishes at a, so some factor in C_q does. The factors in $C(Q)$ that vanish at a therefore form a hitting set, which contains a minimal one H, and $a \in V(H)$. Conversely, let $a \in V(H)$ for some $H \in \mathcal{H}(Q)$. Every $q \in Q$ has a factor in H, which vanishes at a, so $a \in V(Q)$.

Theorem 2 (Termination of refinement). *Algorithm 2 terminates on every finite polynomial system.*

Proof. One round replaces each branch Q by its children $\mathcal{H}(\widetilde{Q})$, where $\widetilde{Q} = \textsc{Interreduce}(Q)$, independently of the other branches. For a finite set G, write $M(G) = \langle \mathrm{LM}(g) : g \in G \rangle$.

We first show that every child H of Q is terminal or satisfies $M(\widetilde{Q}) \subsetneq M(\widetilde{H})$. Every $q \in \widetilde{Q}$ is a multiple of some $h \in H$, so $\mathrm{LM}(h) \mid \mathrm{LM}(q)$ and $M(\widetilde{Q}) \subseteq M(H)$. Suppose that equality holds, and let $h \in H$ divide $q \in \widetilde{Q}$. Since $\mathrm{LM}(h) \in M(\widetilde{Q})$, some $q' \in \widetilde{Q}$ has $\mathrm{LM}(q') \mid \mathrm{LM}(h) \mid \mathrm{LM}(q)$. Since $\widetilde{Q}$ is interreduced, $q' = q$, so q/h has leading monomial 1, and $h = q$ because both are monic. Every element of H is a factor of an element of $\widetilde{Q}$, and every element of $\widetilde{Q}$ has a factor in H, so $H = \widetilde{Q}$. Then H is interreduced and consists of monic irreducible polynomials, so $\widetilde{H} = H$ and $\mathcal{H}(H) = \{H\}$, that is, H is terminal. Otherwise $M(\widetilde{Q}) \subsetneq M(H) \subseteq M(\widetilde{H})$, since interreduction does not shrink the leading-monomial ideal.

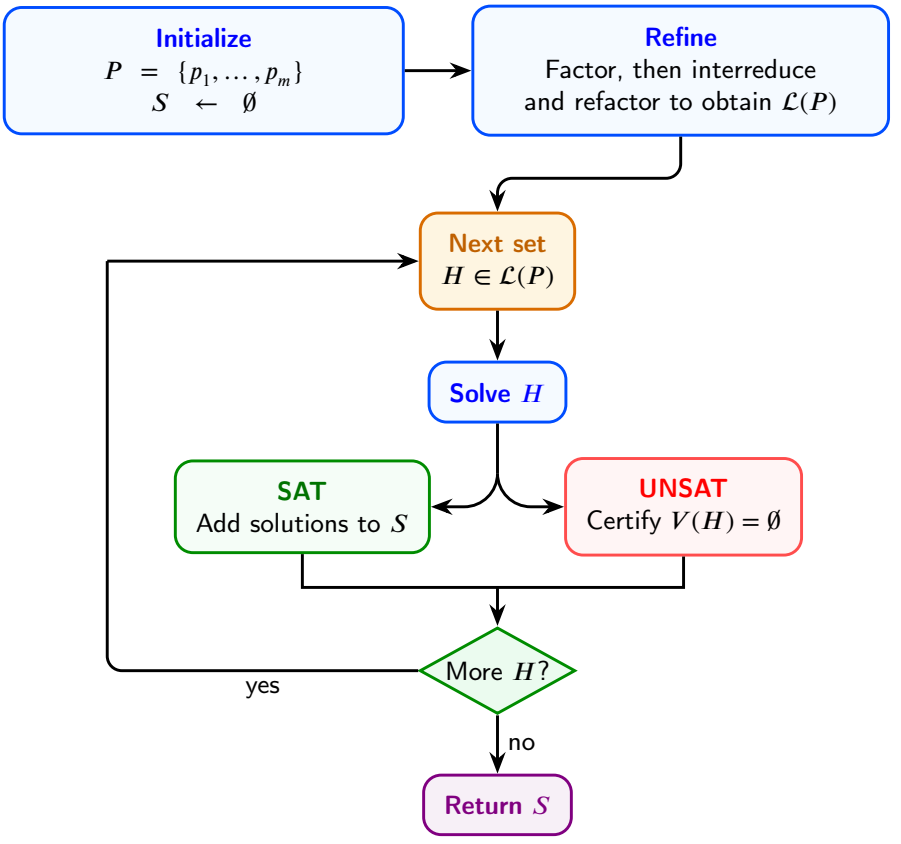


Figure 2: Factor branching procedure of Algorithm 3. Refinement produces terminal branches, which the exact solver either represents or certifies as UNSAT.

Next, we show that the rounds started from $\{Q\}$ reach a fixed point. Since R is Noetherian, it has no infinite strictly ascending chain of ideals, so by Noetherian induction we may assume this for every branch H with $M(\widetilde{Q}) \subsetneq M(\widetilde{H})$. A terminal Q is a fixed point. Otherwise, each of the finitely many children H of Q is terminal or has $M(\widetilde{Q}) \subsetneq M(\widetilde{H})$, so it reaches a fixed point after some k_H rounds, and $\{Q\}$ reaches one after $1 + \max_H k_H$ rounds. Since rounds act on each branch separately, $W = \mathcal{H}(P)$ reaches a fixed point, and the loop stops.

Finally, every branch of the final W is terminal. Otherwise, choose a nonterminal $Q \in W$ whose $M(\widetilde{Q})$ is minimal among the nonterminal branches of W. Since W is a fixed point, Q is a child of some $Q' \in W$. A terminal Q' has only the child Q', so Q' is nonterminal, and $M(\widetilde{Q'}) \subsetneq M(\widetilde{Q})$ contradicts the choice of Q.

Each element of a child divides a different generator of its parent, which makes the child a smaller and simpler system in the following sense.

Theorem 3 (Refined branches are simpler). *Let $H \in \mathcal{H}(\widetilde{Q})$ with $H \neq \widetilde{Q}$, where $\widetilde{Q} = \textsc{Interreduce}(Q)$. Each element of H divides a different element of $\widetilde{Q}$. The leading monomials of H are smaller than those of $\widetilde{Q}$ in the multiset extension of $\prec$, yet generate a strictly larger monomial ideal.*

Proof. Let $h \in H$. Since H is a minimal hitting set, $H \setminus \{h\}$ is not one, so some $q \in \widetilde{Q}$ has $C_q \cap H = \{h\}$. Let $\iota(h)$ be such a q. Then $h \mid \iota(h)$, and ι is injective, since $C_{\iota(h)} \cap H = \{h\}$ determines h. If $h \neq \iota(h)$, then $\iota(h)/h$ is not constant, since both polynomials are monic, so $\mathrm{LM}(h) \prec \mathrm{LM}(h)\,\mathrm{LM}(\iota(h)/h) = \mathrm{LM}(\iota(h))$, since $1 \prec v$ for every monomial $v \neq 1$ (Cox et al., 2015, Ch. 2, §4). The multiset of the leading monomials of H arises from that of $\widetilde{Q}$ by replacing $\mathrm{LM}(\iota(h))$ with $\mathrm{LM}(h)$ for every $h \in H$ and deleting $\mathrm{LM}(q)$ for every $q \in \widetilde{Q} \setminus \iota(H)$. Since $H \neq \widetilde{Q}$, some h has $h \neq \iota(h)$, or some q lies outside $\iota(H)$, so the multiset decreases strictly. Finally, with M as in the proof of Theorem 2, that proof shows that $M(\widetilde{Q}) \subseteq M(H)$, with equality only if $H = \widetilde{Q}$.

Lemma 1 (Refinement preserves solutions). *Every round of Algorithm 2 preserves* $\bigcup_{Q\in W} V(Q)$. *Consequently,*

$$V(P) = \bigcup_{H\in\mathcal{L}(P)} V(H) \qquad \text{and} \qquad \sqrt{\langle P\rangle} = \bigcap_{H\in\mathcal{L}(P)} \sqrt{\langle H\rangle}.$$

PROOF. Interreduction keeps $V(Q)$, so Theorem 1 gives $V(Q) = \bigcup_{H\in\mathcal{H}(\widetilde{Q})} V(H)$ for every branch Q, with $\widetilde{Q} = \textsc{Interreduce}(Q)$, and a round keeps the union. The optional PRUNE removes only branches whose solution sets lie in that of a kept branch. The loop starts from $W = \mathcal{H}(P)$, so Theorem 1 and induction on the rounds give the first identity. By the Nullstellensatz, $\sqrt{\langle P\rangle} = I(V(P))$ and $\sqrt{\langle H\rangle} = I(V(H))$, and the vanishing ideal of a finite union is the intersection of the vanishing ideals, which gives the second.

Theorem 4 (Factor branching). *Algorithm 3 terminates and returns a decomposition of* $V(P)$*:*

$$V(P) = \bigcup_{H\in S} V(H) \qquad \text{and} \qquad \sqrt{\langle P\rangle} = \bigcap_{H\in S} \sqrt{\langle H\rangle}.$$

PROOF. By Theorem 2, $\textsc{Refine}(P)$ returns the finite set $\mathcal{L}(P)$, so the algorithm makes finitely many solver calls, each of which returns. The branches that $\mathcal{B}$ certifies UNSAT have empty solution sets, and $\mathcal{B}$ represents every other branch exactly, so the first identity follows from Lemma 1, and the second follows from the first by the Nullstellensatz, as in the proof of Lemma 1.

3.3. Minimal primes

Let $\mathcal{B}_{\min}$ be a branch solver that returns the minimal primes of its input. If we run Algorithm 3 with $\mathcal{B}_{\min}$ as a branch solver and apply the following postprocessing, we get the minimal primes of $\langle P\rangle$.

The output S is then the set of primes that $\mathcal{B}_{\min}$ returns on the branches in $\mathcal{L}(P)$, and the postprocessing keeps its minimal elements under inclusion in two steps.

1. Represent each prime by its reduced Gröbner basis. Equal primes have the same reduced Gröbner basis, so this merges the copies of a prime that several branches return.
2. Remove every $\mathfrak{p} \in S$ that contains another element $\mathfrak{p}'$ of S. The inclusion $\mathfrak{p}' \subseteq \mathfrak{p}$ holds exactly when every generator of $\mathfrak{p}'$ reduces to zero modulo the Gröbner basis of $\mathfrak{p}$.

A prime that strictly contains $\mathfrak{p}'$ has smaller dimension than $\mathfrak{p}'$, so the second step tests $\mathfrak{p}' \subseteq \mathfrak{p}$ only when $\mathfrak{p}'$ has the larger dimension.

Theorem 5 (Prime factor branching). *With the branch solver* $\mathcal{B}_{\min}$*, the output* S *of Algorithm 3 after the postprocessing is the minimal primes of* $\langle P\rangle$.

PROOF. By Theorem 2, $\mathcal{L}(P)$ is finite, so S is finite. On each $H \in \mathcal{L}(P)$, $\mathcal{B}_{\min}$ returns the minimal primes of $\langle H\rangle$, which intersect to $\sqrt{\langle H\rangle}$, so Lemma 1 gives

$$\bigcap_{\mathfrak{p}\in S} \mathfrak{p} = \bigcap_{H\in\mathcal{L}(P)} \sqrt{\langle H\rangle} = \sqrt{\langle P\rangle}.$$

In particular, every $\mathfrak{p} \in S$ contains $\langle P\rangle$.

Let $\mathfrak{q}$ be a minimal prime of $\langle P\rangle$. Every $\mathfrak{p} \in S$ with $\mathfrak{p} \subseteq \mathfrak{q}$ is a prime containing $\langle P\rangle$, so $\mathfrak{p} = \mathfrak{q}$ by the minimality of $\mathfrak{q}$. Such a $\mathfrak{p}$ exists, because a prime that contains a finite intersection of ideals contains one of them (Atiyah and Macdonald, 1969, Prop. 1.11), and $\mathfrak{q}$ contains $\bigcap_{\mathfrak{p}\in S} \mathfrak{p}$. Hence $\mathfrak{q} \in S$, no other element of S lies in $\mathfrak{q}$, and $\mathfrak{q}$ is a minimal element of S.

Conversely, let $\mathfrak{p}$ be a minimal element of S. It contains $\sqrt{\langle P\rangle}$, the intersection of the finitely many minimal primes of $\langle P\rangle$, so by the same proposition it contains one of them, $\mathfrak{q}$. By the previous paragraph $\mathfrak{q} \in S$, and the minimality of $\mathfrak{p}$ in S gives $\mathfrak{p} = \mathfrak{q}$.

4. Ideal-preserving factor branching

Algorithm 3 discards the exponents $e_{q,f}$ and hence only describes $V(P)$ and $\sqrt{\langle P\rangle}$. But ideal membership, multiplicities, and embedded components depend on $I = \langle P\rangle$ itself. This section adapts factor branching so that it preserves the ideal.

4.1. Ideal reduction

For a finite system Q and $h \in C(Q)$, let $a_h = \max\{e_{q,h} : q \in Q,\ h \in C_q\}$ be the largest exponent of h.

Suppose that a factor h lies in every SAT minimal hitting set of Q. Then h vanishes on all of $V(Q)$, and the Nullstellensatz puts some power of h in $\langle Q\rangle$. Lemma 2 shows that the power h^{a_h} suffices, so the factorization bounds the exponent without a Gröbner basis of $\langle Q\rangle$.

For example, $Q = \{xy,\ xz,\ yz - 1\}$ is interreduced for every monomial order. Its factor sets $\{x, y\}$, $\{x, z\}$, and $\{yz - 1\}$ have the minimal hitting sets $\{x, yz - 1\}$ and $\{y, z, yz - 1\}$. The second is UNSAT, since $y = z = 0$ contradicts $yz - 1 = 0$. Hence x and $yz - 1$ are in every SAT minimal hitting set, $a_x = 1$ and $a_{yz-1} = 1$, and indeed $x^1, (yz - 1)^1 \in \langle Q\rangle$.

Algorithm 4 adds each h^{a_h} to the generators, drops the generators they divide, and repeats on the new system. We say that $\mathcal{B}$ certifies $V(H) = \emptyset$ when Algorithm 3 with $\mathcal{B}$ certifies every terminal branch of H as UNSAT. If no minimal hitting set is SAT, the intersection is $C(Q)$, and Lemma 2 still applies. On the example, the first round replaces xy and xz by x, giving $\{x, yz - 1\}$.

Algorithm 4 Ideal reduction

Require: A finite polynomial system P and a branch solver $\mathcal{B}$
Ensure: A finite generating set Q of $\langle P\rangle$

$Q \leftarrow$ INTERREDUCE(P)
loop
  $\mathcal{H}_{\text{sat}} \leftarrow \{H \in \mathcal{H}(Q) : \mathcal{B}$ does not certify $V(H) = \emptyset\}$
  $\Delta \leftarrow \{h^{a_h} : h \in \bigcap \mathcal{H}_{\text{sat}}\} \setminus Q$
  if $\Delta = \emptyset$ **then**
    return Q
  end if
  $Q \leftarrow$ INTERREDUCE($\Delta \cup \{q \in Q :$ no element of Δ divides $q\}$)
end loop

The output generates $\langle P\rangle$, with polynomials replaced by a factor to the largest exponent that appears in the system, yielding a simpler system.

Each round needs only the intersection of the SAT minimal hitting sets, so an implementation need not classify every minimal hitting set. It keeps the factors F that lie in every SAT hitting set checked so far. Starting from $F = C(Q)$, it skips every hitting set that contains F, intersects F with every other hitting set that is SAT, and stops once F is empty. Each skipped hitting set contains F, so the final F is $\bigcap \mathcal{H}_{\text{sat}}$. Two disjoint SAT hitting sets already empty F, so a round that finds nothing often costs only a few solver calls.

4.2. Primary factor branching

Algorithm 3 with $\mathcal{B}_{\min}$ describes only $\sqrt{I}$. A minimal primary decomposition of I also records the multiplicity of each component and embedded components. Algorithm 5 computes one with an algorithm $\mathcal{B}_{\text{prim}}$ that returns a minimal primary decomposition of its input. Its other steps make $\mathcal{B}_{\text{prim}}$ cheaper, skip it, or label its output.

The algorithm first replaces P by the output Q of Algorithm 4. By Theorem 6, $\langle Q\rangle = I$, so every later step works on a smaller description of I.

It then computes the minimal primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ of I, which give the isolated components in the optional lines and mark the components of $\mathcal{B}_{\text{prim}}$. Algorithm 3 on Q with the branch solver $\mathcal{B}_{\min}$ and the postprocessing of Subsection 3.3 finds them without decomposing I itself, and it can skip the minimal hitting sets of Q that the last round of Algorithm 4 certified as UNSAT.

Optionally, the minimal primes give the isolated components $\mathfrak{q}_1, \ldots, \mathfrak{q}_r$ without $\mathcal{B}_{\text{prim}}$. They intersect to I exactly when I has no embedded components, and the algorithm then returns them, so these lines pay off only on such ideals. Each $\mathfrak{q}_i$ comes from a *separator* of $\mathfrak{p}_i$, a polynomial $w_i \in \bigcap_{j\neq i} \mathfrak{p}_j \setminus \mathfrak{p}_i$, which exists because $\mathfrak{p}_i$ contains no other minimal prime and hence, by Subsection 2.2, not their intersection. Saturating by w_i removes every component whose radical does not contain $\mathfrak{p}_i$ (Shimoyama and Yokoyama, 1996), so $\mathfrak{q}_i$ is the top-dimensional part of $I : w_i^{\infty}$, the intersection of its primary components of maximal dimension (Eisenbud et al., 1992) (Lemma 3). A Gröbner basis of I decides whether $\mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_r = I$.

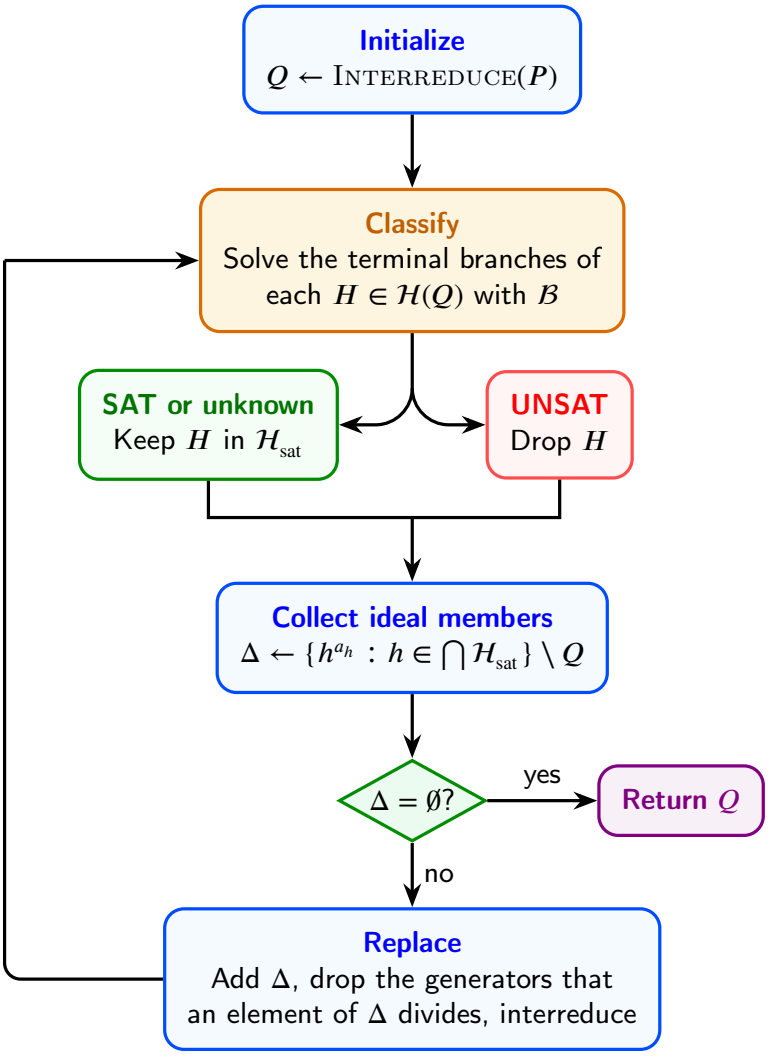


Figure 3: Ideal reduction of Algorithm 4. Every factor h chosen by all SAT minimal hitting sets enters the system as h^{a_h}, which replaces the generators it divides. The loop stops when no new power appears, and $\langle Q \rangle = \langle P \rangle$ throughout.

Otherwise, $\mathcal{B}_{\text{prim}}$ decomposes $\langle Q \rangle$, which is I.

A component $\mathfrak{q}$ is isolated if $\sqrt{\mathfrak{q}}$ is one of the $\mathfrak{p}_i$ and embedded otherwise. The radical of an embedded component strictly contains at least one $\mathfrak{p}_i$, so the variety of the component lies inside an irreducible component of $V(I)$.

For example, $I = \langle x^2, xy \rangle$ has the single minimal prime $\langle x \rangle$, which needs no separator. The top-dimensional part of I is $\langle x \rangle \neq I$, so $\mathcal{B}_{\text{prim}}$ runs and returns $I = \langle x \rangle \cap \langle x^2, y \rangle$. The component $\langle x \rangle$ is isolated, and $\langle x^2, y \rangle$ is embedded, since its radical $\langle x, y \rangle$ strictly contains $\langle x \rangle$.

A separator also needs no intersection of primes. The saturation $I : \mathfrak{p}_i^{\infty} = \{g \in R : \mathfrak{p}_i^m g \subseteq I \text{ for some } m \geq 0\}$ keeps exactly the components whose radicals do not contain $\mathfrak{p}_i$, so it lies in every other minimal prime but not in $\mathfrak{p}_i$, and every element of its Gröbner basis outside $\mathfrak{p}_i$ is a separator.

A zero-dimensional I needs neither separators nor the intersection test: its primes are maximal, so it has no embedded components. Its components are pairwise comaximal, so $\dim_K R/I$ is the sum of the $\dim_K R/\mathfrak{q}_i$, each at least $\dim_K R/\mathfrak{p}_i$ with equality only if $\mathfrak{q}_i = \mathfrak{p}_i$. If $\dim_K R/I = \sum_i \dim_K R/\mathfrak{p}_i$, every component is prime. Otherwise $\mathfrak{q}_i = I + \mathfrak{p}_i^m$ for the first m at which $\dim_K R/(I + \mathfrak{p}_i^m)$ stops growing, by Nakayama's lemma (Atiyah and Macdonald, 1969, Prop. 2.6).

Algorithm 5 Primary factor branching

Require: A finite polynomial system P, branch solvers $\mathcal{B}$ and $\mathcal{B}_{\min}$, and a minimal primary decomposition algorithm $\mathcal{B}_{\text{prim}}$
Ensure: A minimal primary decomposition of $\langle P\rangle$, each component marked isolated or embedded

$Q \leftarrow \textsc{IdealReduce}(P, \mathcal{B})$ ▷ Algorithm 4, $\langle Q\rangle = \langle P\rangle$
$\{\mathfrak{p}_1, \dots, \mathfrak{p}_r\} \leftarrow$ the minimal primes of $\langle Q\rangle$ ▷ Theorem 5
for all $i \in \{1, \dots, r\}$ **do** ▷ Optional
  $w_i \leftarrow$ a separator of $\mathfrak{p}_i$
  $\mathfrak{q}_i \leftarrow$ the top-dimensional part of $\langle Q\rangle : w_i^{\infty}$
end for
if $\mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_r = \langle Q\rangle$ **then**
  return $\{\mathfrak{q}_1, \dots, \mathfrak{q}_r\}$, all marked isolated
end if
$\mathcal{D} \leftarrow \mathcal{B}_{\text{prim}}(Q)$
for all $\mathfrak{q} \in \mathcal{D}$ **do**
  mark $\mathfrak{q}$ isolated if $\sqrt{\mathfrak{q}} \in \{\mathfrak{p}_1, \dots, \mathfrak{p}_r\}$, and embedded otherwise
end for
return $\mathcal{D}$

The mark needs no equality test of ideals. A prime that strictly contains $\mathfrak{p}_i$ has smaller dimension (Subsection 3.3), so $\mathfrak{q}$ is isolated exactly when the generators of some $\mathfrak{p}_i$ of its dimension reduce to zero modulo a Gröbner basis of $\sqrt{\mathfrak{q}}$.

4.3. Theoretical analysis

Lemma 2 (SAT branches). *Let $\mathcal{H}_{\text{sat}} \subseteq \mathcal{H}(Q)$ contain every SAT minimal hitting set, and let $h \in C(Q)$ lie in every $H \in \mathcal{H}_{\text{sat}}$. Then $h^{a_h} \in \langle Q\rangle$.*

Proof. For $q \in Q$ with $h \in C_q$, let $r_q = q/h^{e_{q,h}}$ be the cofactor of h. We first show that the cofactors have no common zero on $V(Q)$. Suppose that $b \in V(Q)$ is a common zero of the cofactors. Then every $q \in Q$ has a factor $f \neq h$ with $f(b) = 0$: a factor of r_q if $h \in C_q$, and any factor that vanishes at b otherwise. So the factors $f \neq h$ in $C(Q)$ that vanish at b form a hitting set, and a minimal hitting set H inside it has $b \in V(H)$ and $h \notin H$. Then H is SAT, hence $H \in \mathcal{H}_{\text{sat}}$ and $h \in H$, a contradiction. By the Nullstellensatz, $1 = \sum_{q\in Q} g_q\, q + \sum_{q:\, h\in C_q} g'_q\, r_q$ for some $g_q, g'_q \in R$. Since $e_{q,h} \le a_h$, every $h^{a_h} r_q = h^{a_h - e_{q,h}} q$ lies in $\langle Q\rangle$, and multiplying by h^{a_h} gives $h^{a_h} \in \langle Q\rangle$.

Theorem 6 (Ideal reduction). *Algorithm 4 terminates and returns a finite generating set Q of $\langle P\rangle$.*

Proof. Every round keeps $\langle Q\rangle$. Indeed, $\mathcal{B}$ certifies $V(H) = \emptyset$ only when it holds (Lemma 1), so $\mathcal{H}_{\text{sat}}$ contains every SAT minimal hitting set. Hence $\Delta \subseteq \langle Q\rangle$ by Lemma 2, the dropped generators are multiples of elements of Δ, and interreduction keeps the ideal.

For termination, let $M(G) = \langle \mathrm{LM}(g) : g \in G\rangle$ as in the proof of Theorem 2. Consider a round with $\Delta \neq \emptyset$, and let $Q_0 = \Delta \cup \{q \in Q : \text{no element of } \Delta \text{ divides } q\}$, so that the round ends with $\textsc{Interreduce}(Q_0)$. At the start of every round, Q is interreduced, and in particular monic. Every $q \in Q$ is kept or is a multiple of an element of Δ, so $M(Q) \subseteq M(Q_0)$. Let $\delta = h^{a_h} \in \Delta$ and let $q \in Q$ have $e_{q,h} = a_h$, so that $\delta \mid q$. If $\mathrm{LM}(\delta) \in M(Q)$, some $q' \in Q$ has $\mathrm{LM}(q') \mid \mathrm{LM}(\delta) \mid \mathrm{LM}(q)$, so $q' = q$ since Q is interreduced. Then q/δ has leading monomial 1, and $\delta = q \in Q$ since both are monic, which contradicts $\delta \notin Q$. Hence $M(Q) \subsetneq M(Q_0)$, and since interreduction does not shrink the leading-monomial ideal, $M(Q)$ grows strictly in every round that does not return. Since R is Noetherian, there are finitely many rounds, and each makes finitely many calls of $\mathcal{B}$.

Lemma 3 (Isolated components by separators). *Let $\mathfrak{p}$ be a minimal prime of an ideal I and w a separator of $\mathfrak{p}$. Then $\mathfrak{p}$ is the only minimal prime of $I : w^{\infty}$, and the $\mathfrak{p}$-primary component of I is the intersection of the primary components of $I : w^{\infty}$ of maximal dimension.*

Proof. Let $I = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_t$ be a minimal primary decomposition with $\mathfrak{p}_j = \sqrt{\mathfrak{q}_j}$, and let $T = \{j : w \notin \mathfrak{p}_j\}$. By Subsection 2.2, $I : w^{\infty} = \bigcap_{j\in T} \mathfrak{q}_j$, and this decomposition is again minimal, since a $\mathfrak{q}_j$ that contained the intersection

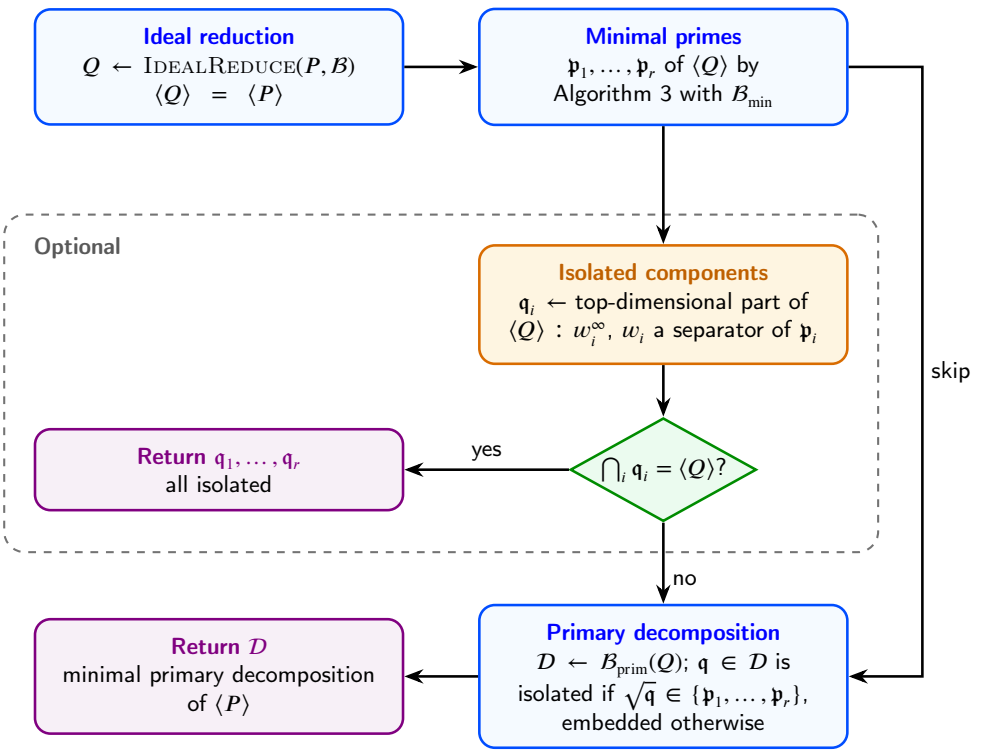


Figure 4: Primary factor branching of Algorithm 5. Ideal reduction shrinks the system to Q without changing the ideal. The separators of the minimal primes give the isolated components, which are returned if they intersect to $\langle Q \rangle$. Otherwise the general solver $\mathcal{B}_{\text{prim}}$ decomposes Q, and the minimal primes mark each component isolated or embedded.

of the other kept components would contain the smaller intersection of all other components. For $j \in T$, the prime $\mathfrak{p}_j$ contains I, hence $\sqrt{I}$, the intersection of the minimal primes of I, and hence one of them (Atiyah and Macdonald, 1969, Prop. 1.11). This minimal prime is $\mathfrak{p}$, since every other one contains w. The $\mathfrak{p}$-primary component $\mathfrak{q}$ of I is kept, since $w \notin \mathfrak{p}$, so $\mathfrak{p}$ contains $I : w^\infty$. Every prime that contains $I : w^\infty$ contains some kept $\mathfrak{q}_j$ (Atiyah and Macdonald, 1969, Prop. 1.11), hence $\mathfrak{p}_j \supseteq \mathfrak{p}$, so $\mathfrak{p}$ is the only minimal prime of $I : w^\infty$.

It follows that $\mathfrak{q}$ is the isolated $\mathfrak{p}$-primary component of $I : w^\infty$, which is the same in every minimal primary decomposition of $I : w^\infty$. Every other component of such a decomposition has a radical that contains $I : w^\infty$, hence $\mathfrak{p}$, and differs from $\mathfrak{p}$. A prime that strictly contains $\mathfrak{p}$ has smaller dimension (Subsection 3.3), so $\mathfrak{q}$ is the only component of maximal dimension.

Theorem 7 (Primary factor branching). *Algorithm 5, with or without its optional lines, terminates and returns a minimal primary decomposition $\mathcal{D}$ of $\langle P \rangle$:*

$$\langle P \rangle = \bigcap_{\mathfrak{q} \in D} \mathfrak{q}.$$

It marks a component isolated exactly when the component is an isolated component of $\langle P \rangle$, and the radical of every embedded component strictly contains a minimal prime of $\langle P \rangle$. If every component is given by a Gröbner basis, then $g \in \langle P \rangle$ if and only if g reduces to zero modulo the basis of every component.

Proof. By Theorem 6, Algorithm 4 terminates and returns Q with $\langle Q \rangle = \langle P \rangle = I$, so by Theorem 5 the $\mathfrak{p}_i$ are the minimal primes of I. The remaining lines make finitely many solver calls, so the algorithm terminates.

Suppose that the optional lines return. By Lemma 3, each $\mathfrak{q}_i$ is the isolated $\mathfrak{p}_i$-primary component of I, and $\mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_r = I$. The radicals $\mathfrak{p}_i$ are distinct, and no $\mathfrak{q}_i$ is redundant. Otherwise $\mathfrak{p}_i$ contains $\bigcap_{j \neq i} \mathfrak{q}_j$, hence some $\mathfrak{q}_j$ with $j \neq i$ (Atiyah and Macdonald, 1969, Prop. 1.11), and hence $\mathfrak{p}_j$, which contradicts the minimality of $\mathfrak{p}_i$. So the $\mathfrak{q}_i$ form a minimal primary decomposition of I, and all of them are isolated.

Otherwise, $\mathcal{B}_{\text{prim}}$ decomposes $\langle Q \rangle = I$. A component is isolated exactly when its radical is a minimal prime of I, that is, one of the $\mathfrak{p}_i$, which is the mark. The radical of an embedded component is a prime that contains I, hence some $\mathfrak{p}_i$ as in the proof of Lemma 3, and it differs from $\mathfrak{p}_i$.

In both cases, $g \in \langle P \rangle$ if and only if g lies in every component, and a Gröbner basis decides membership by reduction (Subsection 2.3).

5. Experiments

5.1. Setup

We implemented Algorithm 3 in Rust 1.98.1 with Symbolica 3.0.1 (Ruijl, 2026). Refinement prunes as in Subsection 3.1, and the branch solver computes a Gröbner basis of each terminal branch, fewest generators first, with M4GB, F5, GVW, F4, or Singular 4.4.1 (Decker et al., 2025; Faugère, 1999, 2002; Gao et al., 2016; Makarim and Stevens, 2017); a basis containing 1 certifies UNSAT. The baselines are six whole-ideal Gröbner-basis configurations and three decomposition methods from Singular and Macaulay2 1.26.06 (Decker et al., 2025; Grayson and Stillman, 2026). Every entry is one run on an Apple M4 Pro (14 cores, 24 GB) with a 60-second limit. For factor branching, T is the wall time and F the time from the start of the hitting-set search to the first SAT branch. The measurements are in the `results` directory of the repository listed under Data availability.

5.2. Baseline systems

The baseline has 20 generated systems in six families (Table 1): signed-distance cycles and affine-factor hypergraphs with many cheap branches, overlapping random multiples whose dense quadratic factors recur across equations, and gated systems, which multiply every equation of Cyclic, Katsura, or a dense quadratic system by a new variable (Björck and Fröberg, 1991; Böge et al., 1986) and so have one easy selector branch and one nonlinear residual branch.

Family	Tested sizes	Variables/equations; hitting sets
Signed-distance cycle	$n = 12, 14, 16$	$12/12, 14/14, 16/16$; 924, 3432, 12870
Overlapping random multiples	$(n, p, k) = (4, 6, 2), (6, 8, 3), (8, 10, 3)$	n/n; 4, 16, 28
Positive affine hypergraph	$(n, m, k) = (8, 6, 4), (10, 8, 5), (12, 10, 6)$	n/m; 63, 239, 833
Gated Cyclic	orders $5, 6, 7$	$6/5, 7/6, 8/7$; 2 each
Gated Katsura and dense quadratics	Katsura $6, 7, 8$; dense $n = 5, 6, 7$	$8/7, 9/8, 10/9, 6/5, 7/6, 8/7$; 2 each
Square affine hypergraph	$(n, k) = (8, 4), (10, 5)$	n/n; 178, 309

Table 1
The generated systems used for the baseline benchmark.

Table 2 lists the fastest completed method of each class; "timeout" means that every method of the class timed out. Factor branching completes 19 of the 20 systems and is fastest on 16, against 9 and 16 completed systems for the direct and decomposition classes.

Factor branching pays off when the branch ideals are far easier than the input ideal. Many hitting sets are affordable when each branch is cheap, as in the positive affine family, but one branch as hard as the whole system, as in gated Cyclic-7, removes the gain.

F shows this: it stays at a few milliseconds on positive affine $(12, 10, 6)$ despite 833 hitting sets, and on gated Katsura-8, whose nonlinear residual dominates T.

5.3. Size-graded families

The following families inherit their factors from the problem, and we grow each until every method times out.

- The reduced cyclic n-roots systems `Singular.rcyclic_n`, a variant of the cyclic n-roots problem (Björck and Fröberg, 1991), taken unchanged from SymbolicData[1].
- Nash equilibria of n-player games with two pure strategies per player (Sturmfels, 2002). Each player i either plays a pure strategy, $p_i(p_i - 1) = 0$, or is indifferent, so its payoff difference vanishes. The payoffs are random integers in $[-9, 9]$ (generic) or in $\{-1, 0, 1\}$ (degenerate).

[1] `https://github.com/symbolicdata/data`, commit `a5cdf53`.

System	Best factor branching T/F	Best direct GB	Best decomposition
Signed distance, $n = 12$	F5 2.669/0.006	**F4 0.443**	primdec 3.056
Signed distance, $n = 14$	GVW 9.824/0.006	**Singular 7.421**	primdec 29.894
Signed distance, $n = 16$	**M4GB 37.391/0.006**	timeout	timeout
Overlapping $(4, 6, 2)$	**F5 0.014/0.003**	F4 0.429	facstd 0.265
Overlapping $(6, 8, 3)$	**F5 0.160/0.006**	timeout	timeout
Overlapping $(8, 10, 3)$	**Singular 2.671/0.013**	timeout	timeout
Positive affine $(8, 6, 4)$	**M4GB 0.180/0.003**	timeout	facstd 0.368
Positive affine $(10, 8, 5)$	**F4 0.703/0.003**	timeout	facstd 1.836
Positive affine $(12, 10, 6)$	**F4 2.398/0.003**	timeout	facstd 53.318
Gated Cyclic-5	F4 0.011/0.003	**GVW 0.009**	facstd 0.181
Gated Cyclic-6	**Singular 0.043/0.010**	Singular 0.309	M2 2.620
Gated Cyclic-7	timeout	timeout	timeout
Gated Katsura-6	**Singular 0.047/0.010**	GVW 0.226	facstd 0.233
Gated Katsura-7	**Singular 0.283/0.010**	GVW 3.176	facstd 0.539
Gated Katsura-8	**Singular 2.575/0.010**	timeout	facstd 3.596
Gated dense, $n = 5$	**Singular 0.025/0.010**	M4GB 0.171	facstd 0.198
Gated dense, $n = 6$	**Singular 0.552/0.010**	M4GB 4.736	facstd 0.755
Gated dense, $n = 7$	**Singular 42.670/0.010**	timeout	facstd 48.820
Square affine $(8, 4)$	**M4GB 0.532/0.003**	timeout	facstd 0.601
Square affine $(10, 5)$	**F4 0.920/0.003**	timeout	facstd 2.141
#systems solved	19	9	16
#fastest (ties counted)	16	3	0

Table 2
Fastest completed method of each class on the generated systems, with a 60-second limit; the fastest overall is in bold. Factor branching entries give the wall time T and the time F to the first SAT branch. All times are in seconds.

- Critical points of a random quadratic program f on every face of the box $[0, 1]^n$, given by $x_i(x_i - 1)\,\partial_i f = 0$.
- Equilibria of a random Lotka–Volterra community on every species support (Hofbauer and Sigmund, 1998), given by $x_i\big(r_i + \sum_j a_{ij}x_j\big) = 0$ with $a_{ii} < 0$.

Each size is one seeded instance. Factor branching entries give T only, the direct class adds Singular's `std` in degrevlex (Singular dp) and drops standalone F4, and a method that times out is not run at larger sizes.

In every family of Table 3, the baselines win only on systems they finish in under 0.5 seconds. On the larger systems of each family, factor branching is fastest, and its lead grows with n, to 372 times on rcyclic-18. Only factor branching completes rcyclic-19, which refines into 3 terminal branches far easier than the system, and box-QP with $n = 8$, which refines into 6534 cheap branches.

Factor branching can also use a decomposition method as its branch solver. Table 4 compares `facstd` as a backend against `facstd` alone, Algorithm 5 against primdec, and Algorithm 3 with $\mathcal{B}_{\min}$ against `minAssGTZ`. Here $\mathcal{B}_{\min}$ computes a factorizing Gröbner basis (Gräbe, 1995), returns each linear piece as a prime, and decomposes the other pieces with `minAssGTZ` (Gianni et al., 1988), and only positive-dimensional primes prune later branches. Algorithm 5 uses the factorizing Gröbner basis in Algorithm 4, $\mathcal{B}_{\min}$ for the minimal primes, and the separators and zero-dimensional shortcut of Subsection 4.2, so it runs primdec only when an embedded component exists. In the last two pairs, refinement uses lex order and every Gröbner basis degrevlex.

Factor branching with `facstd` completes every rcyclic system in under 0.3 seconds, the Nash games with $n = 5$, and Lotka–Volterra with $n = 12$; of these, `facstd` alone finishes only rcyclic-8. `facstd` alone wins on box-QP, the $n = 4$ Nash games, and the smaller Lotka–Volterra systems, where factor branching pays for starting Singular once per branch.

Algorithm 3 with $\mathcal{B}_{\min}$ returns the minimal primes of larger systems than `minAssGTZ` alone in every family. On the largest box-QP and Lotka–Volterra systems, it costs at most about 1.5 times as much as factor branching with

System	Best factor branching T	Best direct GB	Best decomposition
rcyclic-8	F4 0.019	**GVW 0.008**	facstd 0.425
rcyclic-10	**F4 0.018**	M4GB 0.041	timeout
rcyclic-12	**F4 0.021**	M4GB 0.198	timeout
rcyclic-14	**F4 0.029**	Singular dp 0.676	timeout
rcyclic-16	**Singular 0.062**	Singular dp 7.154	timeout
rcyclic-17	**Singular 0.091**	M2 19.76	timeout
rcyclic-18	**Singular 0.093**	M2 34.60	timeout
rcyclic-19	**Singular 0.113**	timeout	timeout
Nash generic, $n = 3$	F5 0.044	**F5 0.006**	facstd 0.197
Nash generic, $n = 4$	**GVW 0.136**	Singular dp 0.262	facstd 0.262
Nash generic, $n = 5$	**Singular 1.382**	timeout	timeout
Nash generic, $n = 6$	timeout	timeout	timeout
Nash degenerate, $n = 3$	M4GB 0.043	**GVW 0.005**	facstd 0.179
Nash degenerate, $n = 4$	**F5 0.086**	Singular lex 0.215	facstd 0.203
Nash degenerate, $n = 5$	**Singular 0.714**	timeout	timeout
Nash degenerate, $n = 6$	timeout	timeout	timeout
Box-QP, $n = 4$	F4 0.237	**F5 0.005**	facstd 0.179
Box-QP, $n = 5$	F5 0.700	**Singular dp 0.216**	facstd 0.297
Box-QP, $n = 6$	M4GB 2.174	**F5 0.033**	primdec 0.939
Box-QP, $n = 7$	**F5 6.494**	timeout	facstd 12.63
Box-QP, $n = 8$	**F5 20.42**	timeout	timeout
Box-QP, $n = 9$	timeout	timeout	timeout
Lotka–Volterra, $n = 6$	M4GB 0.191	**F5 0.006**	facstd 0.216
Lotka–Volterra, $n = 8$	F4 0.809	Singular dp 0.701	**facstd 0.464**
Lotka–Volterra, $n = 10$	**GVW 3.002**	timeout	facstd 6.432
Lotka–Volterra, $n = 12$	**F5 13.15**	timeout	timeout
Lotka–Volterra, $n = 14$	timeout	timeout	timeout
#systems solved	27	20	13
#fastest (ties counted)	18	8	1

Table 3
Best completed results across solver classes on the size-graded families, with a 60-second limit. For rcyclic, only selected sizes are shown; the summary rows count all 31 systems, rcyclic-8 to rcyclic-19 included. All times are in seconds.

`std` in the same harness. It stops when `minAssGTZ` cannot decompose one terminal branch, such as the branch of the degenerate Nash game with $n = 5$ on which every player is indifferent.

Algorithm 5 returns the same components as primdec wherever both finish. On box-QP with $n = 5$ and $n = 6$ and Lotka–Volterra with $n = 8$, the separators find no embedded component, so primdec never runs; there it is faster, up to 23 times on box-QP with $n = 5$, and the only one to finish Lotka–Volterra with $n = 8$. With embedded components it pays for both, as on the degenerate Nash game with $n = 4$, and on rcyclic-8 it finds the three isolated components in 0.37 seconds before primdec times out. Algorithm 4 stops after one round on these systems, so the reduced system is the interreduced input. Algorithm 5 needs a Gröbner basis of the ideal and times out wherever `std` does, and on the generic Nash game with $n = 4$ its isolated components do not finish although `std` does.

5.4. Physical system

The physical system is an order-five approximation that constrains graviton couplings in asymptotic-safety models to remain consistent with quantum field theory (QFT) (Rafie-Zinedine, 2018; Vacca and Zanusso, 2010); solving it helps determine admissible fixed-point structures in quantum gravity (Vacca and Zanusso, 2010). It has 28 equations in 21 variables over $\mathbb{Q}(\pi)$, and κ_h must be nonzero. Factor branching and the Singular and Macaulay2 baselines saturate by κ_h; the standalone M4GB, F5, GVW, and F4 runs add the Rabinowitsch generator $1 - s\kappa_h$ (Rabinowitsch, 1930).

	facstd		primdec		minAssGTZ	
System	FB	alone	Primary	alone	Prime	alone
rcyclic-8	**0.038**	0.425	timeout	timeout	**0.081**	10.64
rcyclic-10	**0.040**	timeout	timeout	timeout	**0.110**	timeout
rcyclic-12	**0.043**	timeout	timeout	timeout	**1.472**	timeout
rcyclic-14	**0.053**	timeout	timeout	timeout	**21.22**	timeout
rcyclic-16	**0.079**	timeout	timeout	timeout	timeout	timeout
rcyclic-17	**0.119**	timeout	timeout	timeout	timeout	timeout
rcyclic-18	**0.162**	timeout	timeout	timeout	timeout	timeout
rcyclic-19	**0.285**	timeout	timeout	timeout	timeout	timeout
Nash generic, $n = 3$	**0.138**	0.197	0.065	**0.063**	**0.063**	0.073
Nash generic, $n = 4$	0.449	**0.262**	timeout	timeout	**0.121**	timeout
Nash generic, $n = 5$	**13.48**	timeout	timeout	timeout	timeout	timeout
Nash degenerate, $n = 3$	**0.136**	0.179	0.087	**0.074**	**0.060**	0.069
Nash degenerate, $n = 4$	0.263	**0.203**	6.117	**5.004**	**0.088**	timeout
Nash degenerate, $n = 5$	**1.634**	timeout	timeout	timeout	timeout	timeout
Box-QP, $n = 4$	0.661	**0.179**	**0.089**	0.128	**0.089**	0.170
Box-QP, $n = 5$	1.928	**0.297**	**0.715**	16.20	**0.166**	17.00
Box-QP, $n = 6$	7.528	**1.259**	**0.423**	3.172	**0.446**	1.564
Box-QP, $n = 7$	22.77	**12.63**	timeout	timeout	**1.564**	timeout
Box-QP, $n = 8$	timeout	timeout	timeout	timeout	**5.742**	timeout
Box-QP, $n = 9$	timeout	timeout	timeout	timeout	**23.05**	timeout
Lotka–Volterra, $n = 6$	0.615	**0.216**	**0.101**	0.128	**0.096**	0.161
Lotka–Volterra, $n = 8$	3.012	**0.464**	**1.873**	timeout	**0.254**	timeout
Lotka–Volterra, $n = 10$	10.05	**6.432**	timeout	timeout	**1.059**	timeout
Lotka–Volterra, $n = 12$	**41.47**	timeout	timeout	timeout	**5.286**	timeout
Lotka–Volterra, $n = 14$	timeout	timeout	timeout	timeout	**31.61**	timeout

Table 4
`facstd` as the factor branching (FB) backend and alone, and each new algorithm against the Singular command with the same output, with a 60-second limit; the faster of each pair is in bold. Primary is Algorithm 5, and Prime is Algorithm 3 with $\mathcal{B}_{\min}$. The last two pairs run in one Python process that drives one Singular process and exclude the input conversion, so their times are not comparable with the `facstd` columns or with Table 3. A method that timed out at a smaller size of a family is listed as a timeout. Sizes at which every method times out are omitted, and for rcyclic only the sizes of Table 3 are shown. All times are in seconds.

Every run on the physical systems has a 120-second limit, or 300 seconds in Table 5, runs concurrently with the others, and reports its internal solver time. Singular and Macaulay2 compute over $\mathbb{Q}(\pi)$, except Macaulay2's `decompose`, which cannot, and the Rust programs, which take π as a variable. Every saturation uses Singular's `sat` or Macaulay2's `saturate`, every primary decomposition `primdecSY` (Shimoyama and Yokoyama, 1996), and Algorithm 5 runs its separators. On the physical system, saturations and decompositions use lex order with the min-fill variable order, a greedy elimination order computed from the variable supports of the equations, as refinement and Algorithm 4 always do. Factor branching limits its saturation of the system to five seconds, removes κ_h from every factor set, drops a branch once κ_h reduces to zero modulo its generators, and adds $1 - s\kappa_h$ to each terminal branch.

All nine standalone baselines time out on the physical system, whereas factor branching, with every backend, refines the saturated system into 10 SAT terminal branches and solves them in 0.87 to 0.94 seconds, about 0.83 seconds of which go to the saturation by κ_h.

Keeping κ_h nonzero changes only the SAT test of Algorithm 4: a minimal hitting set H counts as SAT when $V(H) \not\subseteq V(\kappa_h)$. The proof of Lemma 2 then shows that Q and the cofactors of h have no common zero outside $V(\kappa_h)$,

so the Nullstellensatz gives $\kappa_h^s h^{a_h} \in \langle Q \rangle$ for some $s \geq 0$. Each round then keeps $\langle Q \rangle : \kappa_h^\infty$ instead of $\langle Q \rangle$, and the termination argument of Theorem 6 is unchanged. Algorithm 5 then decomposes $\langle Q \rangle : \kappa_h^\infty$, whose minimal primes are those of $\langle Q \rangle$ that do not contain κ_h.

Without a time limit, Algorithm 5 returns a minimal primary decomposition of the physical system over $\mathbb{Q}(\pi)$. Algorithm 4, keeping κ_h nonzero, reduces the 28 equations to 18 generators in three rounds and 0.6 seconds; Algorithm 3 with $\mathcal{B}_{\min}$ finds the six minimal primes in 0.1 seconds; the separators give their isolated components in 1.6 seconds, but these do not intersect to the ideal; and `primdecSY` decomposes the saturation, with 56 generators, in most of the time of Primary in Table 5, whereas on the unreduced system with a Rabinowitsch generator it did not finish in 30 minutes. Six of the ten components are isolated, in two families of dimensions 13, 12, and 8: prime in one family, and in the other obtained from the radical by squaring one generator, a variable, which doubles the degree. Each family also has two embedded components, of dimension 10 inside its 12-dimensional component and of dimension 8 inside its 13-dimensional one, and a Gröbner basis of the saturation confirms that the components intersect to $\langle P \rangle : \kappa_h^\infty$. With the variables in alphabetical order instead, Algorithm 4 stops at 27 generators, because the factorizing Gröbner basis of the largest hitting set does not finish within 120 seconds.

For each $\alpha \in \{1, 1/2, 1/4, 2, 4\}$, we also derive two systems from the Einstein–Hilbert truncation of gravity coupled to fermions and scalars, each with 31 equations once the Rabinowitsch equation of `KappaRefFinal` is replaced by saturation. The graviton-only system sets $d_1 = d_2 = d_3 = 0$ (29 or 30 variables), and the fermion-only system sets every variable whose name begins with Dp or p to zero (15 variables). On these systems, saturations and decompositions use degrevlex with the variables in name order, which is faster there. Table 5 runs the pairs of Table 4 on them and on the physical system.

	`facstd`		primdec		`minAssGTZ`	
System	FB	alone	Primary	alone	Prime	alone
Physical	**0.954**	timeout	**271.3**	timeout	**0.962**	timeout
Graviton only, $\alpha = 1$	timeout	timeout	timeout	timeout	timeout	timeout
Graviton only, $\alpha = 1/2$	timeout	timeout	timeout	timeout	timeout	timeout
Graviton only, $\alpha = 1/4$	timeout	timeout	timeout	timeout	timeout	timeout
Graviton only, $\alpha = 2$	timeout	timeout	timeout	timeout	timeout	timeout
Graviton only, $\alpha = 4$	timeout	timeout	timeout	timeout	timeout	timeout
Fermion only, $\alpha = 1$	0.035	**0.001**	0.059	**0.007**	0.267	**0.007**
Fermion only, $\alpha = 1/2$	0.028	**0.001**	0.060	**0.007**	0.229	**0.007**
Fermion only, $\alpha = 1/4$	0.037	**0.001**	0.060	**0.006**	0.231	**0.007**
Fermion only, $\alpha = 2$	0.028	$<$ **0.001**	0.060	**0.007**	0.231	**0.007**
Fermion only, $\alpha = 4$	0.031	**0.001**	0.060	**0.006**	0.240	**0.007**

Table 5
The pairs of Table 4 on the physical systems; the faster of each pair is in bold. Every method keeps κ_h, or `KappaRefFinal`, nonzero: the commands alone saturate first, factor branching as described above, Primary by Algorithm 4 and a saturation of its output, and Prime by dropping the minimal primes that contain the coupling. Times are internal solver times in seconds, with a 300-second limit.

No method finishes a graviton-only system. On the fermion-only systems, every method finishes within 0.3 seconds with the same single prime and primary component, and the Singular commands alone win every pair; factor branching refines these systems into one branch. On the physical system, every command alone times out, whereas factor branching with `facstd` and Prime finish in under a second and Primary in 271 seconds.

6. Conclusion

We introduced a solver-agnostic framework that decomposes polynomial systems into smaller branches by enumerating minimal hitting sets of their factor sets.

We proved that refinement terminates and preserves the full solution set, and that the variants return the minimal primes, a generating set of the ideal, and a minimal primary decomposition.

On the baseline benchmarks, factor branching completed 19 of 20 systems, compared with 9 for direct Gröbner-basis methods and 16 for decomposition methods, and on the size-graded families 27 of 31, with a lead that grew with the system size. On the QFT system, where one coupling must be nonzero, imposing this by saturation lets factor branching complete with all six backends in under one second, while all nine standalone baselines time out. Primary factor branching decomposes the same saturated system in under five minutes.

Future work will test this benefit on other systems, and find additional ways to break down polynomial systems into smaller branches which apply to a wider range of systems.

CRediT authorship contribution statement

Ilay Menahem: Conceptualization, Methodology, Software, Validation, Formal analysis, Data curation, Writing – original draft, Visualization. **Michael Shalyt:** Writing – review & editing. **Ido Kaminer:** Writing – review & editing, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The code and data associated with this work are publicly available at https://github.com/IlayMenahem/soothing.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used generative AI tools to assist with drafting and editing portions of the manuscript text. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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